Lab 7: HPGe Detectors: Efficiency Calibration and spectroscopy

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Abstract

Gamma spectroscopy is a widely used technique to characterize both the composition and activity of radioactive material. In this laboratory, we characterize the (energy-dependent) detection efficiency of a High Purity Germanium (HPGe) detector, and use this detector to estimate the activity of a ^{154}Eu source of unknown age. We measured gamma spectra of ^{137}Cs, ^{54}Mn, ^{60}Co, ^{57}Co, and ^{133}Ba sources with known initial activity and age, of the ^{154}Eu source with known initial activity but unknown age, and of background radiation, and identified the count rates of key photopeaks at 1332, 1173, 835, 662, 356, 302, 276, 136, 133, and 81 keV, comparing using known activity and branching factors to build an intrinsic efficiency curve of the HPGe detector. Using this curve, we estimated detection efficiency at ^{154}Eu peaks of 244, 344, and 779 keV to compare with branching factors to estimate the current activity and probable age of the source. We found an efficiency curve matching initial expectations, having a hump of high efficiency at low energy with steadily decreasing efficiency for higher energies. Using this curve, we estimated the activity of the ^{154}Eu source to be 2.56954 ± 0.03766 μCi, compared to a known initial activity of 10 μCi, implying the age of the ^{154}Eu source to be 26.499 ± 0.286 years, with a manufacturing date around September 1999. The results demonstrate the consistency of the model of efficiency as a combination of intrinsic and geometric factors, and demonstrate the effectiveness of gamma spectroscopy (across consistent geometric layouts) as a tool to measure the activity and age of radioisotope samples. They may be of interest to those aiming to use spectroscopy to date objects using radioactive material, and in particular to those seeking to use HPGe scintillators for this purpose.

1 Introduction and Background

Gamma spectroscopy is a crucial technique for characterizing radioactive material. While the unique spectra created as a result of rich interaction phenomena from the different combinations of radiation released by radioisotopes serve well to distinguish and identify such isotopes, the intensity (count rates) of known features are also numerically useful. Most notably, since the rate of gamma emissions at a specific energy can be predicted from isotope properties and a source's overall activity, the number of counts a detector registers near that energy can be used to estimate a *detection efficiency* of the spectrometry system at that energy, and thus predict the activity of an identified radioisotope from the spectrum the detector produces in response to a source. This efficiency is energy dependent, and the relationship may be quantified for a given detector and geometry.

In this laboratory, we perform gamma spectrum measurements of several sources of known composition and activity using a High Purity Germanium (HPGe) scintillator, to characterize the energy-dependent efficiency of this detector and predict the age of a ^{154}Eu source with known initial activity.

1.1 Detection Efficiency

The overall, or "absolute" efficiency ε_{abs} of a detector, that is the ratio of decay instances detected N to the known true count S, may be broken up into an "intrinsic" efficiency to

the detector ε_{intr}, and a "geometric" efficiency (a.k.a. fractional solid angle) Ω_f of the given experimental setup:

$$\varepsilon_{abs} = \frac{N}{S} = \Omega_f \varepsilon_{intr} \tag{1}$$

A solid angle Ω (units of steradians, $0 \leq \Omega \leq 4\pi$) describes a field of view, in this case the angular exposure of the detector to the source. The fractional solid angle $\Omega_f = \Omega/4\pi$, describing the fraction of radiation quanta which might even reach the detector in a given setup, can be calculated for the specific geometry of that setup, and used to estimate the *intrinsic* efficiency of the detector at that energy, which can be applied to other measurements at that energy. By taking efficiency estimates over a representative range of energies, the efficiency may be estimated at other energies (such as by using a polynomial fit).

1.2 Specific Activity

While the overall activity of a source might be known, this must be then interpreted using decay properties to find the activity expected for a specific type of radiation, such as the 1408 keV gamma rays emitted by a ^{154}Eu sample. Thus, the true specific activity S for a source of known activity A measured over a sufficiently short time T may be described using Eq. 2:

$$S = BF \times BR \times T \times A \tag{2}$$

In Eq. 2, BR represents the *branching ratio*, the fraction of all decays which are of the specific decay mode associated with this energy (such as the fraction of decays which are electron capture, as opposed to the competing beta). BF then represents the *branching fraction*, the fraction of decays in that mode which release the desired form of radiation (such as the fraction which then emit 662 keV gamma rays specifically). These values have been measured and tabulated across a wide variety of isotopes and decay phenomena, and may be readily used for these specific activity and efficiency estimates. For an isotope of half-life $t_{1/2}$ (and corresponding activity coefficient $\lambda = \ln(1/2)/t_{1/2}$), the age t of a sample can be used to estimate the activity A from its initial activity A_0, or vice versa, using the exponential decay rate law[1]:

$$A = A_0 e^{-\lambda t} \tag{3}$$

1.3 Solid Angle Estimation

The solid angle "visible" to the detector can be calculated by integrating over the visible area of the detector from the perspective of the source location[2]:

$$\Omega = 4\pi\Omega_f = \int_A^0 \frac{\cos\alpha}{r^2} dA \tag{4}$$

In the conceptual integral of Eq. 4, For a detector of substantial volume, this must be extended over that volume (or taken over an average, representative area), and likewise this integral must be nested and evaluated over the volume of a non-point source.
This is equivalent to projecting the area of the detector onto a normalized sphere. In the

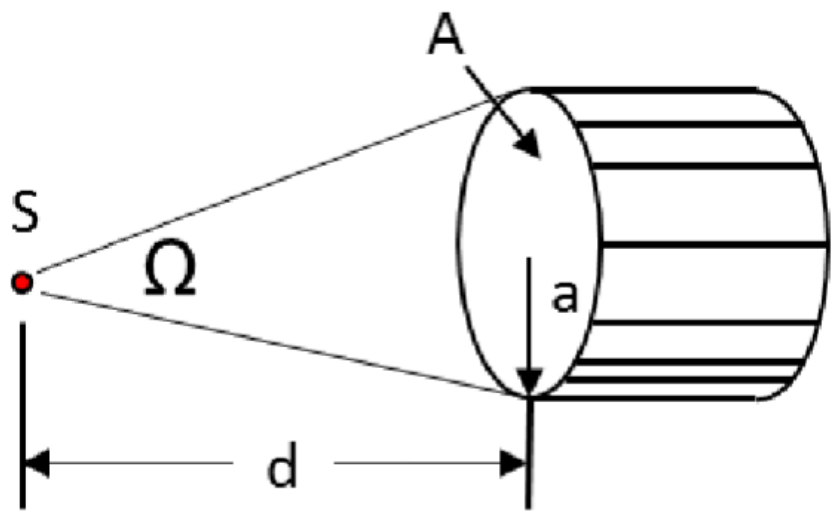


Figure 1: A geometric approximation, of the source as a point aligned with a perfectly cylindrical detector with radius a at distance d, to estimate the solid angle subtended by the detector.

case of a sufficiently small source, aligned to the axis of a circular detector (as shown in Fig. 1), the circle already overlaps with a portion of a sphere with radius $R = \sqrt{a^2 + d^2}$, and so this integral can be evaluated by comparing that region to the overall sphere radius:

$$\begin{aligned} \Omega_f = \frac{\Omega}{4\pi} = \frac{A}{4\pi R^2} &= \frac{1}{4\pi R^2} \int_0^{2\pi} d\phi \int_0^{\arctan(r/d)} R^2 \sin\theta d\theta \\ \Omega_f &= \frac{2\pi R^2}{4\pi R^2} \left[-\cos\theta\right]\Big|_{\theta=0}^{\arctan(r/d)} \\ \Omega_f &= \frac{1}{2}\left(1 - \cos\left[\arctan(r/d)\right]\right) \\ \Omega_f &= \frac{1}{2}\left(1 - \frac{d}{\sqrt{r^2 + d^2}}\right) \end{aligned} \tag{5}$$

To extend Eq. 5 to a volumetric, cylindrical detector rather than a perfectly flat circle, an average distance $\langle d \rangle$ to circular cross-sections of the cylinder (of radius r, height h, and a face distance c away from the source) may be substituted[3]:

$$\langle d \rangle = \frac{r^2}{2} + \frac{h^2}{12} + c^2 \tag{6}$$

Given known dimensions, and an error in distance σ_d, the error of this estimate can also be determined:

$$\sigma(f(x, y, \dots)) = \sqrt{(\frac{df}{dx}\sigma_x)^2 + (\frac{df}{dy}\sigma_y)^2 \dots} \tag{7}$$

$$\sigma(f(x,y,\dots)) = \sqrt{(\frac{df}{dx}\sigma_x)^2 + (\frac{df}{dy}\sigma_y)^2 \dots}$$
$$\sigma(\Omega_f(d)) = \sqrt{(\frac{d\Omega_f}{dd}\sigma_d)^2} \tag{8}$$
$$\sigma(\Omega_f(d)) = \Omega_f\left(\frac{2\sigma_d}{d}\right)$$

where, in Eq. 8, if d is replaced with an average distance $\langle d \rangle$, the error is likewise replaced with $\sigma_{\langle d \rangle} = \frac{d\langle d \rangle}{dc}\sigma_c$ for a distance to front c, as in Eq. 6.

As many detectors are of cylindrical shape[2], these approximations can be highly useful in radiation detection, provided proper care is taken to align the source on the axis of the cylinder, and a sufficiently small source (or one at sufficiently large distance) is used.

1.4 High Purity Germanium Scintillators

High Purity Germanium (HPGe) scintillators are a class of semiconductor detectors with high energy resolution compared to other similar scintillators, and are thus extremely useful in spectrometry measurements such as of ^{133}Ba, which release gamma rays of several closely-packed energies. Tsoulfanidis and Landseberger provide a more detailed explanation of HPGe detectors and their usage[1], but for the purposes of this laboratory, it is most crucial that HPGe detectors, in addition to having high energy resolution, may be stored at room temperature, requiring active cooling only during usage for best performance.

2 Experimental Procedure

We arranged an HPGe scintillator, cooling device, and multichannel analyzer (MCA) to measure activity from a coaxially located radiation source, as shown in Fig. 2. The MCA was connected to a workstation computer, and was set using the computer to provide a bias of +3400 V. Crucially, the cooling device was active during all measurements to cool the HPGe detector to usable temperatures. Dimensions of the HPGe detector, alongside other material details, may be found in Appendix A. We placed a marked foam block against the detector, such that the source would be placed 45 ± 1 mm from the detector.

We measured six sources of known composition (^{137}Cs, ^{54}Mn, ^{60}Co, ^{57}Co, ^{133}Ba, and ^{154}Eu) and activity, alongside standalone background radiation, in order to estimate the age of the ^{154}Eu using the other five sources of known age. For each measurement, we placed the source (if any) in the foam slot 45 ± 1 mm away from the detector (that is, 50 ± 1 mm from the internal crystal), corresponding to a geometric efficiency (or fractional solid angle) of 0.04492 ± 0.00134 using Eqs. 5 and 8 (adjusting both solid angle and error calculations for the length of the detector).

Each spectrum measurement consisted of placing the source (if applicable) in the foam fixture at the fixed distance aligned to the detector, after which we used the computer to ini-

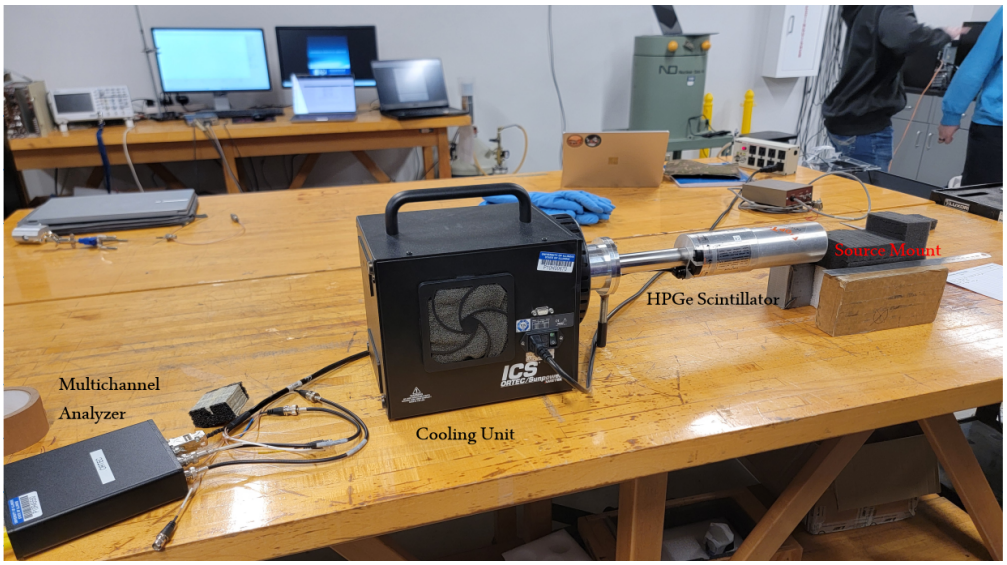
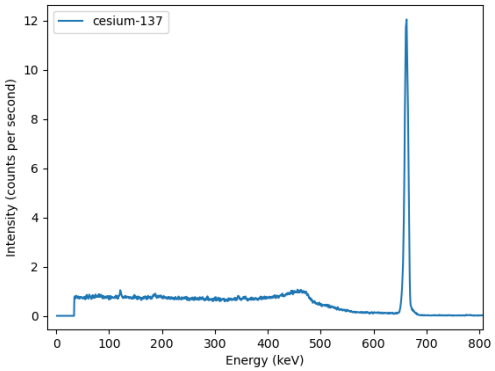


Figure 2: The HPGe detector setup used in this laboratory.

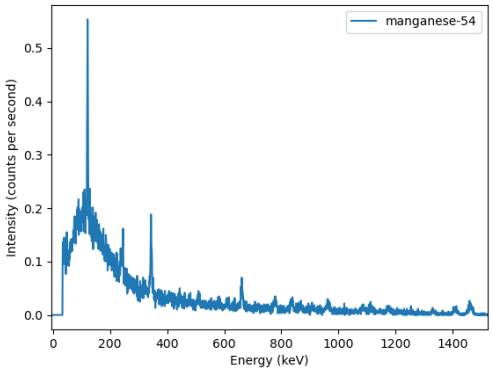
tiate a spectrum recording over a set time (10 minutes each for the five fully known sources, 5 minutes for the ^{154}Eu source, and 20 minutes for background with no source), before swapping out for the next source, taking care not to disturb the foam mount.

3 Experimental Results

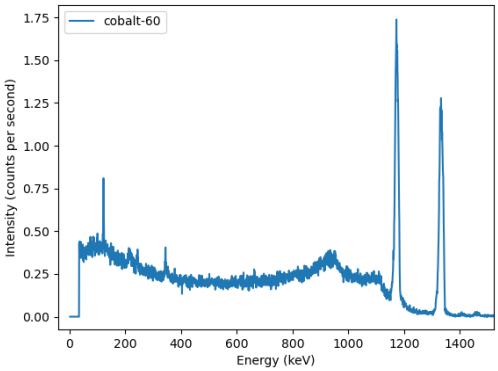
We measured gamma spectra for the six sources and background, which may each be seen in Figs. 3 for the five fully known sources, for the ^{154}Eu source in Fig. 4, and for the background in Fig. 5.



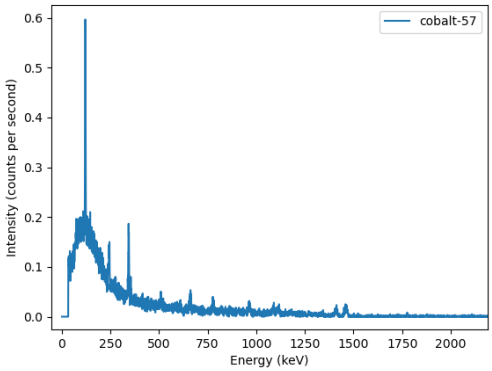
(a) Of the ^{137}Cs source.



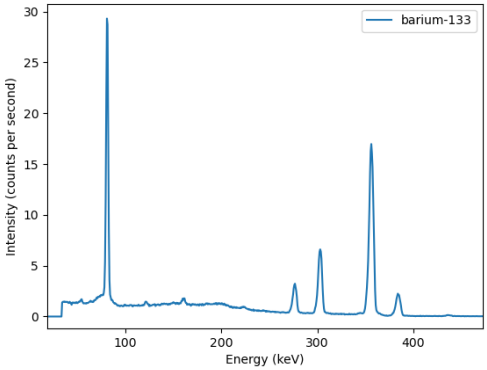
(b) Of the ^{54}Mn source.



(c) Of the ^{60}Co source.



(d) Of the ^{57}Co source.



(e) Of the ^{133}Ba source.

Figure 3: Gamma spectra recorded of the ^{137}Cs, ^{54}Mn, ^{60}Co, ^{57}Co, and ^{133}Ba sources, all over a period of 10 minutes.

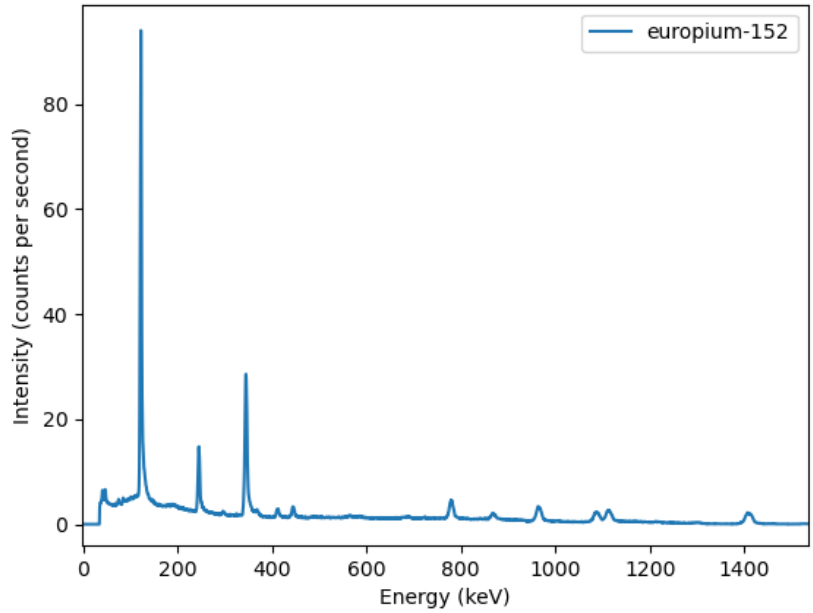


Figure 4: The gamma spectrum recorded of the ^{154}Eu source over 5 minutes.

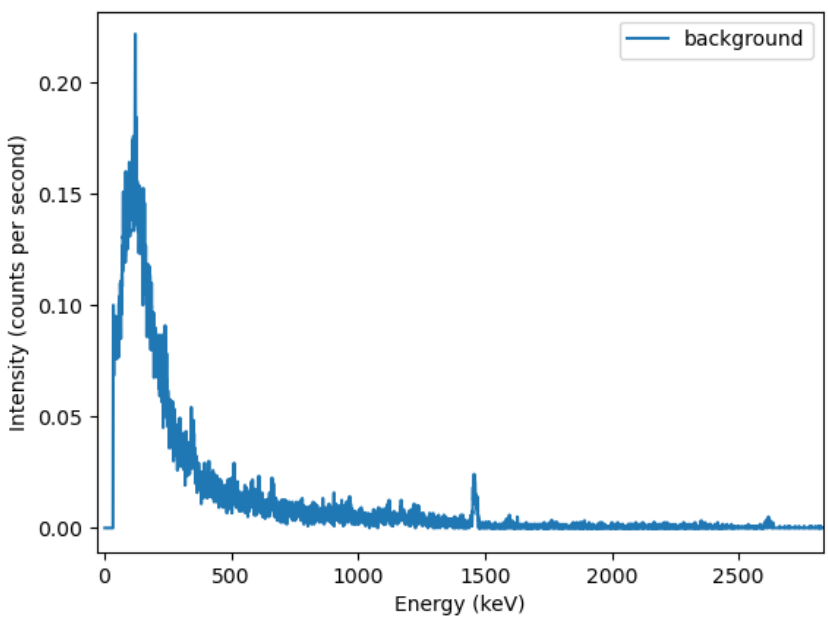


Figure 5: The background gamma spectrum, as recorded over 20 minutes.

Using the clearly identifiable peaks of 661.7 keV in the ^{137}Cs spectrum and 1173 keV in the ^{60}Co spectrum, we found a calibration relationship $E = 0.65662n + 1.63249$ keV, where

Table 1: Peak information, both from observed spectra and expected phenomena, and the resulting intrinsic efficiency calculated at their corresponding energies, for the photopeaks chosen in this experiment. For all five known sources, measurements were taken over a time $T = 600$ seconds.

Isotope	E_0 (keV)	Total N (counts)	BR	BF	$t_{1/2}$ (y)	A (μCi)	ε_{intr}
^{137}Cs	662	93502 ± 476	1.0	0.851	30.080	0.821	0.1342 ± 0.0041
^{60}Co	1173	21207 ± 412	1.0	0.998	5.271	0.324	0.0657 ± 0.0023
^{60}Co	1332	19193 ± 245	1.0	1.000	5.271	0.324	0.0594 ± 0.0019
^{54}Mn	835	186 ± 28	1.0	1.000	0.855	0.001	0.1815 ± 0.0279
^{133}Ba	81	68831 ± 472	1.0	0.329	10.551	0.570	0.3683 ± 0.0113
^{133}Ba	276	10915 ± 144	1.0	0.072	10.551	0.570	0.2684 ± 0.0088
^{133}Ba	302	26014 ± 198	1.0	0.183	10.551	0.570	0.2497 ± 0.0077
^{133}Ba	356	82174 ± 316	1.0	0.620	10.551	0.570	0.2331 ± 0.0070
^{133}Ba	384	10868 ± 121	1.0	0.089	10.551	0.570	0.2140 ± 0.0068
^{57}Co	136	175 ± 27	1.0	0.107	0.744	0.000	4.4535 ± 0.6999
^{57}Co	122	1102 ± 41	1.0	0.856	0.744	0.000	3.4990 ± 0.1670

n is the channel number of the spectrum. We used this relationship to quantify the energy of all spectra used in this laboratory.

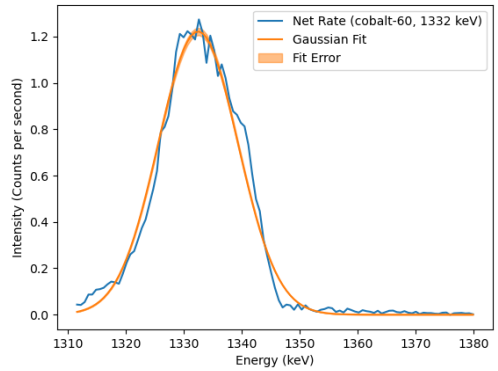
Using the multichannel analysis software Maestro[4] (by converting to an appropriate format, as elaborated upon in Appendix B), we fit Gaussian peaks to the spectrum peaks at 1332, 1173, 835, 662, 356, 302, 276, 136, 133, and 81 keV.

These peak fits may be seen in Figs. 6a through 7d. Using these peaks, the expected activity of each source given ages and half lives[5], tabulated branching ratios and branching factors on the table of nuclides[5], and the geometric efficiency of the detector (0.04492 ± 0.00134), we estimated the intrinsic efficiency at each peak energy, tabulated in Table 1.

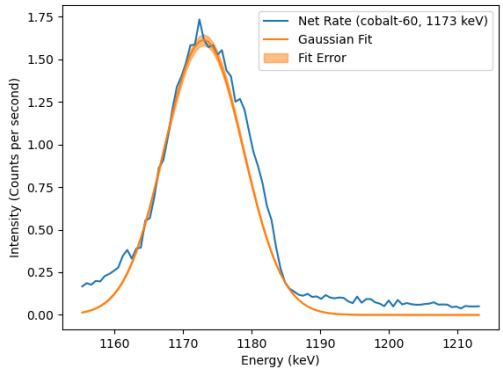
Performing an error-weighted, four-term polynomial fit of $\ln(\varepsilon_{intr})$ versus $\ln(E)$ (such as described by Eq. 9), we obtained a four-term logarithmic polynomial fit of intrinsic efficiency at each different energy.

$$\begin{aligned} y &\approx ax^3 + bx^2 + cx + d \\ y &= \ln(\varepsilon_{intr}) \\ x &= \ln(E) \end{aligned} \tag{9}$$

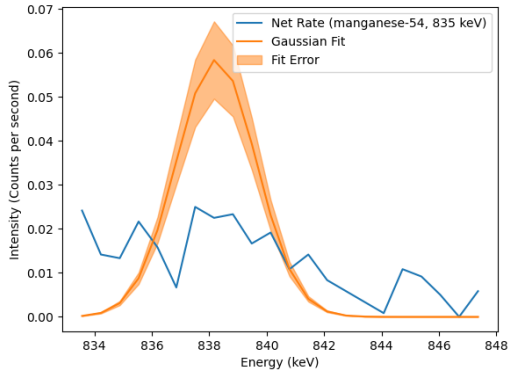
The fit taken over the ^{133}Ba, ^{60}Co, and ^{137}Cs measurements may be viewed in Fig. 8, with a closer view in Fig. 9, while a less usable fit additionally containing the negligible ^{57}Co and ^{54}Mn may be seen in Fig. 10.



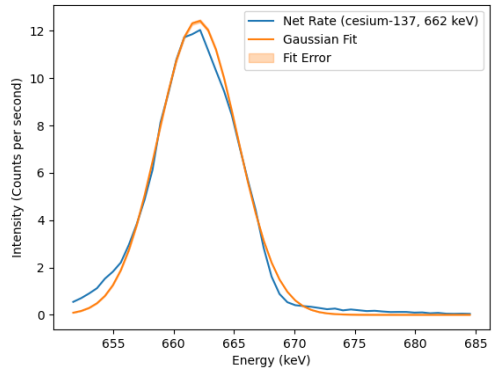
(a) 1332 keV.



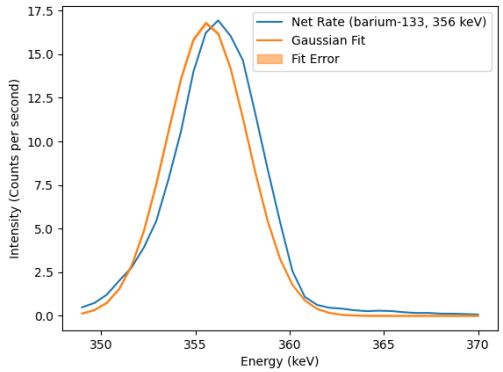
(b) 1173 keV.



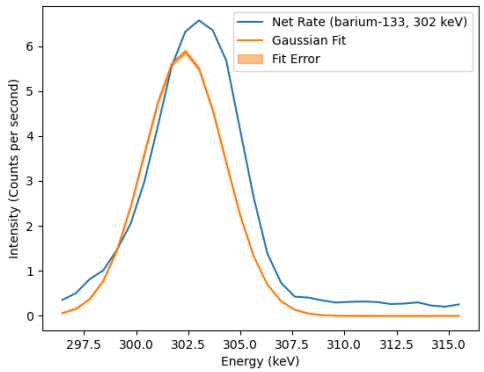
(c) 835 keV.



(d) 662 keV.

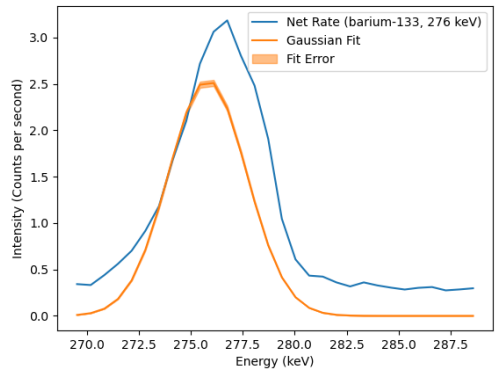


(e) 356 keV.

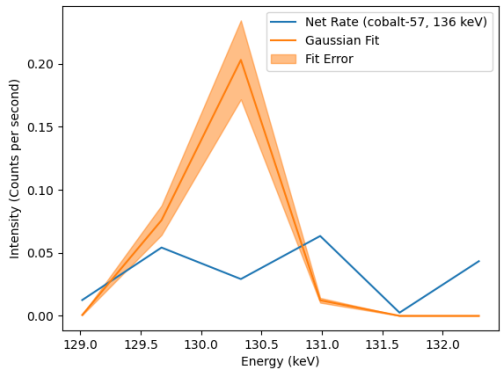


(f) 302 keV.

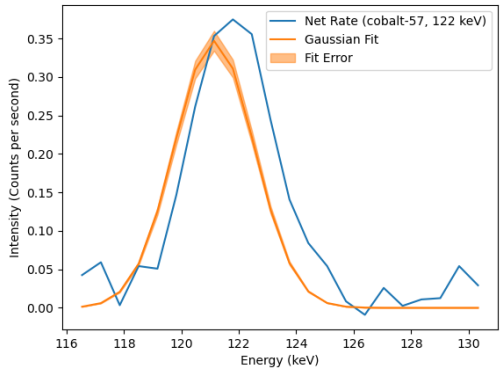
Figure 6: Spectra measured near expected peaks of 1332, 1173, 835, 662, 356, and 302 keV, compared to Gaussian fits of these features.



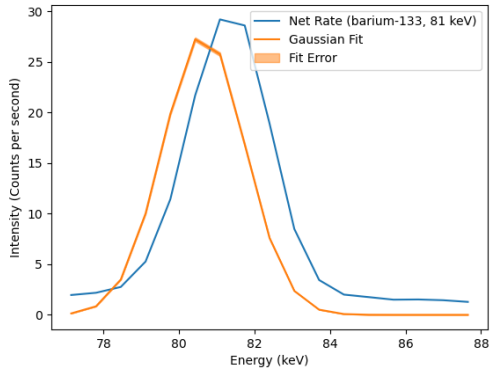
(a) 276 keV.



(b) 136 keV.



(c) 122 keV.



(d) 81 keV.

Figure 7: Spectra measured near expected peaks of 276, 136, 122, and 81 keV, compared to Gaussian fits of these features.

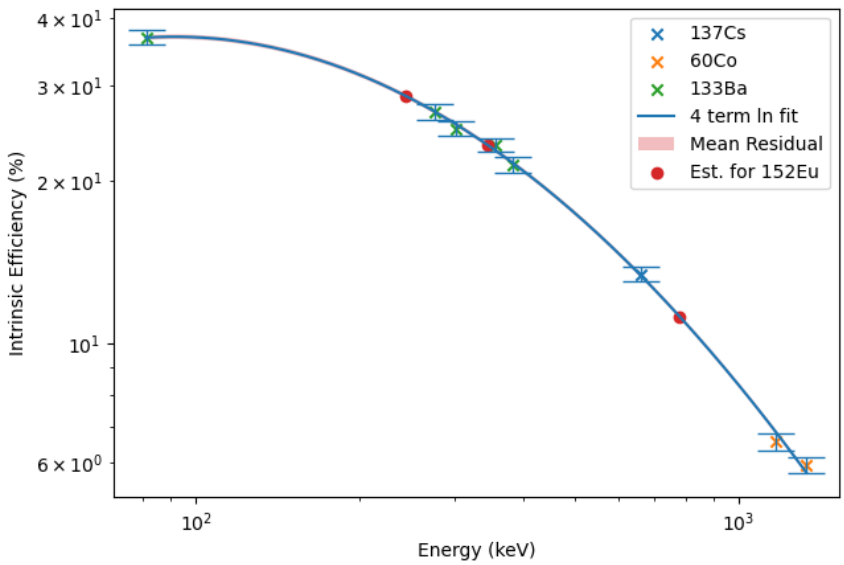


Figure 8: A four-term weighted fit of $\ln(\varepsilon_{intr})$ versus $\ln(E)$ for the detector measurements, for sources with usable activity (excluding ^{57}Co and ^{54}Mn).

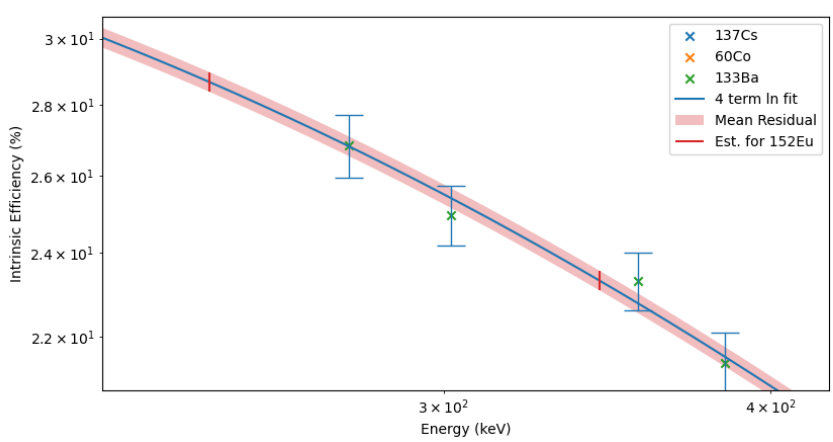


Figure 9: A selected portion of the weighted fit of $\ln(\varepsilon_{intr})$ versus $\ln(E)$ for the detector measurements, for sources with usable activity (excluding ^{57}Co and ^{54}Mn), zoomed in to highlight the slim residuals of activities from those predicted by the fit.

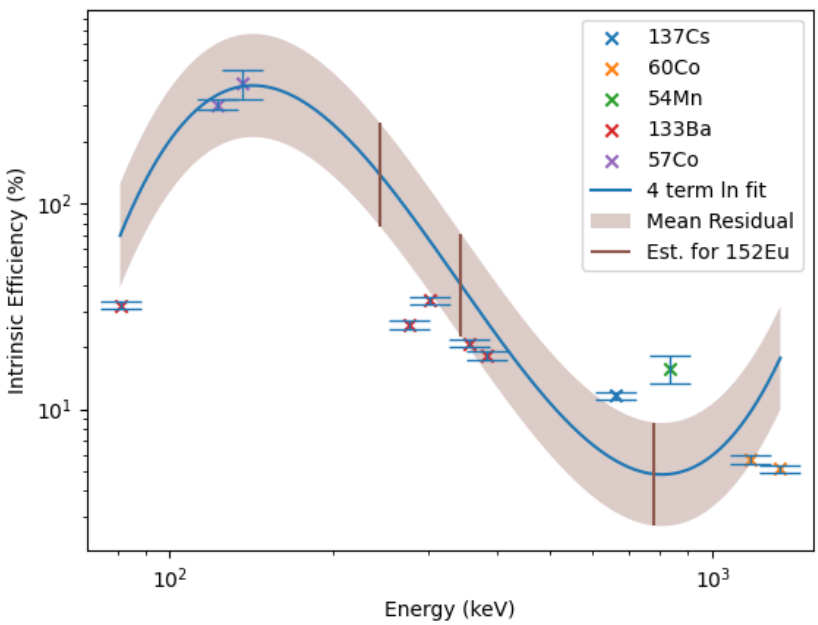


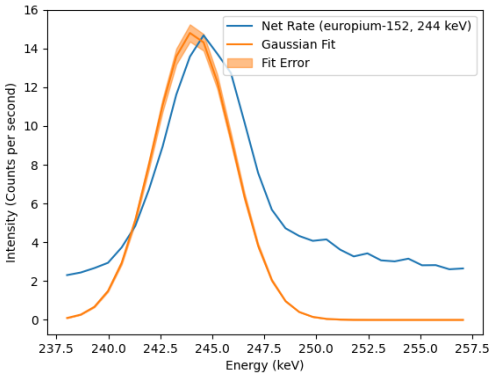
Figure 10: A four-term weighted fit of $\ln(\varepsilon_{intr})$ versus $\ln(E)$ for the detector measurements, including sources ^{57}Co and ^{54}Mn with negligible activity.

We then fit Gaussian peaks to estimate counts at peaks of 244, 344, and 779 keV in the ^{154}Eu spectrum (which may be observed in Figs. 11a through 11c). We selected these three peaks because of their prominence and energies falling comfortably within the range used to estimate intrinsic efficiency, as displayed in the fit in Fig. 8. Using tabulated branching data[5], the fit of ε_{intr}, the geometric efficiency, and Eqs. 1 and 2, we produced estimates of the current activity of the ^{154}Eu source using each of these three peaks, tabulated in Table 2.

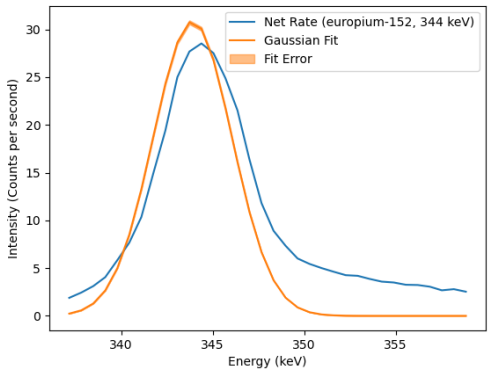
Using these three peaks (and propagating error using Eq. 7 where applicable), we estimated an activity of the ^{154}Eu sample to have been 2.56954 ± 0.03766 μCi. Given the known half-life of ^{154}Eu, 13.52 years[5], and an original activity of 10.0 μCi [3], we estimated using Eq. 3 a sample age of 26.499 ± 0.286 years. Compared to the laboratory date of March 27, 2026, this yields an estimated creation date of approximately September 1999.

Table 2: The peak intensities, tabulated branching information, and corresponding estimates of the ^{154}Eu source, of photopeaks at 244, 344, and 779 keV. This measurement was taken over 5 minutes.

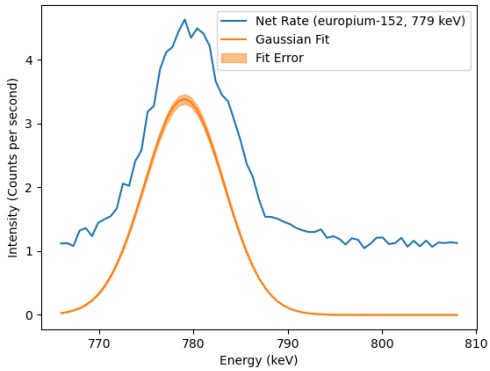
E (keV)	Total N (counts)	Est. ε_{abs}	Est. A (μCi)
244	32375 ± 968	0.01288 ± 0.00013	2.98547 ± 0.09433
344	76021 ± 492	0.01048 ± 0.00011	2.46667 ± 0.02983
779	16301 ± 357	0.00503 ± 0.00005	2.25649 ± 0.05453



(a) 244 keV.



(b) 344 keV.



(c) 779 keV.

Figure 11: Spectra measurements near expected peaks in ^{154}Eu spectra, of 244, 344, and 779 keV, compared to Gaussian fits of these features.

4 Discussion

As the detector used in this experiment was of non-negligible length, an average distance was used as shown in Eq. 6. As such, the error in the measurement of distance $c = 50 \pm 1$ mm to the tip of the detector had to be propagated twice, first through the average distance (Eq. 6), then through the solid angle calculation of Eq. 5:

$$
\begin{aligned}
\sigma(f(x, y, \dots)) &= \sqrt{(\frac{df}{dx}\sigma_x)^2 + (\frac{df}{dy}\sigma_y)^2 \dots} \\
\sigma(f(x)) &= \left|\frac{df}{dx}\sigma_x\right| \\
\sigma(\Omega_f(c)) &= \left|\frac{d\Omega_f}{dc}\sigma_c\right| \\
&= \left|\frac{d\Omega_f}{d\langle d\rangle}\frac{d\langle d\rangle}{dc}\sigma_c\right| \\
&\rightarrow \frac{d\Omega_f}{d\langle d\rangle} = -\frac{1}{2}\left(\frac{a^2}{(\langle d\rangle^2 + a^2)^{1.5}}\right) \\
&\rightarrow \frac{d\langle d\rangle}{dc} = \frac{c}{\sqrt{c^2 + \frac{a^2}{2} + \frac{h^2}{12}}} \\
&= \left|-\frac{1}{2}\left(\frac{a^2}{(\langle d\rangle^2 + a^2)^{1.5}}\right)\frac{c}{\sqrt{c^2 + \frac{a^2}{2} + \frac{h^2}{12}}}\sigma_c\right| \\
\sigma(\Omega_f(c)) &= \frac{1}{2}\left(\frac{a^2}{\left(\sqrt{\frac{3a^2}{2} + \frac{h^2}{12} + c^2}\right)^3}\right)\frac{c}{\sqrt{c^2 + \frac{a^2}{2} + \frac{h^2}{12}}}\sigma_c
\end{aligned}
\tag{10}
$$

We find that the logarithmic fit of intrinsic efficiency to energy (Fig. 8) fairly closely matched that expected from previous experiments[3], having a region of high efficiency near low energies before gradually becoming lower and lower for higher energies. Additionally, as seen most clearly in Fig. 9, the fit captures the measured efficiency at each peak within the margin of error for that peak's efficiency. However, this requires the elimination of the peaks from ^{57}Co and ^{54}Mn: as seen in Table 1, and Figs. 6c and 7b, these isotopes possess such short half-lives that the samples' activity could not be feasibly separated even from the noise of the spectrum (even after subtracting background activity using the 20 minutes background recording), yielding results without meaning (such as a > 1 efficiency estimate using the ^{57}Co sample). Were we to include these uninterpretable "data", an unusable efficiency fit, seen in Fig. 10 is found, yielding a much less useful age estimate of 30.67217 ± 8.80292 years.

Using the appropriate intrinsic efficiency fit, the estimates for activity of the ^{154}Eu source stayed relatively consistent across the three peaks (Table 2), which allow us using the known original activity of the source (10 μCi), to estimate that the sample is 26.499 ± 0.286 years

old, created around September of 1999.

While we carry out error analysis accounting for errors such as in peak fitting or measurement of detector distance, we recognize that other sources of error may have affected the laboratory. These include potential epistemic errors, such as falsely observing the ^{57}Co and ^{54}Mn activities to be negligible, or the polynomial fit being inappropriate (or of too high degree). Additionally, even after subtracting background radiation, some amount of background noise can still be seen, possibly owing to time variation in background radiation, or simply random statistical phenomena. For example, a 2204 keV peak, still visible in some spectra after background subtraction, is a typical indicator of ^{222}Rn decay (through the daughter product ^{214}Bi)[6].

5 Conclusions

The objectives of this laboratory, which aimed to build an efficiency curve of an HPGe detector, and use this curve to estimate the unknown activity and age of a radioactive sample, were largely met.

Spectra were measured of ^{137}Cs, ^{54}Mn, ^{60}Co, ^{57}Co, and ^{133}Ba sources for 10 minutes, of an ^{154}Eu source for 5 minutes, and of background radiation for 20 minutes.

A logarithmic four-term polynomial fit was created of intrinsic efficiency ε_{intr} as a function of energy E for the HPGe scintillator using the sources of known age (all except ^{154}Eu), matching expected behavior with highest efficiency at a range of lower energies, and decreasing efficiency over higher energies.

This curve was used to estimate the current activity of the ^{154}Eu at 2.56954 ± 0.03766 μCi, which, compared to the known original activity of 10.0 μCi, yielded an estimated manufacturing date of September 1999.

References

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A Equipment Details

In order to better replicate experiments and identify issues, the details of each equipment piece are recorded in Table 3.

Further details of equipment are listed in Table 3. Where a detail could not be identified for a given device, it is replaced with "N/A".

Table 3: Details of the equipment used in this laboratory.

Item	Manufacturer	Model	ID Type	ID Number
Multichannel Analyzer	Ortec[a]	EasyMCA	Inventory #	P10R45550
HPGe Scintillator	Ortec[a]	GEM10P4	Serial #	44-TP21936B
^{137}Cs Source	Spectrum[b]	1.0 μCi	Manuf.	Sep 1, 2017
^{60}Co Source	"	"	Manuf.	Sep 1, 2017
^{57}Co Source	"	"	Manuf.	Oct 1, 2017
^{133}Ba Source	"	"	Manuf.	Sep 1, 2017
^{54}Mn Source	"	"	Manuf.	Oct 1, 2017
^{154}Eu Source	Amersham[c]	10 μCi	Manuf.	Unknown
Computer	Dell[d]	Precision 360	Eng. IT ID #	EWS 20426

a Advanced Measurement Technology: 801 South Illinois Avenue, Oak Ridge, Tennessee 37830, United States; Phone: +1.865.482.4411; Fax: +1.865.483.0396; https://www.ortec-online.com; Email: ortec.info@ametek.com

b Spectrum Techniques: 106 Union Valley Rd, Oak Ridge, TN 37830; Phone: 865.482.9937; https://www.spectrumtechniques.com/

c Amersham plc (Acquired by GE Healthcare); 500 West Monroe Street, Chicago, Illinois, 60661; Phone: 1 203 3604369; https://www.gehealthcare.com/

d Dell Technologies: Dell Technologies 1 Dell Way Round Rock, TX 78664.; Phone: 1-877-275-3355; https://www.dell.com/en-us/lp/contact-us

B Additional Codes

To assist in reliably evaluating derivatives of functions for error propagation, the tool `Desmos` was used, in a repository viewable at https://www.desmos.com/calculator/t2e4wghni1.

Additionally, to open spectrum data in the software `Maestro`, a `Python` script was used, which may be viewed at https://github.com/osanstrong/NPRE451/blob/main/report7/make_maestro_specs.py.